

Executive Decision Paper & Capital Allocation Memo

Doc Ref: UHG-CAP-2026-089	Owning Dept: Product & Utilization Mgmt	Target Budget: \$1.2M CapEx / \$180k OpEx	Status: APPROVED FOR Q2 EXECUTION
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1. Strategic Executive Summary & BLUF

Bottom Line Up Front (BLUF): Requesting \$1,200,000 CapEx authorization to deploy the *OptumAuth-AI* Prior Authorization Engine across 40 high-volume outpatient surgical and oncology procedures. Project will generate **\$1,850,000 net annualized operational savings** per region, reduce turnaround time from 72 hours to 8.2 minutes, and guarantee 100% compliance with federal CMS-0057-F interoperability mandates.

2. Business Problem & Financial Impact Model

Strategic Metric	Legacy Baseline	Target SLA	Measured Production	Financial Impact
Turnaround Time (TAT)	72.0 Hours	< 15.0 Mins	8.2 Mins	Eliminates 18% care delays / therapy abandonment
Review Nurse Touch-Time	28.5 min / case	< 5.0 min / case	3.4 min / case	Reallocates 13 FTE nurses (\$1,025,544/yr labor savings)
First-Pass Clean Approval	74.2%	> 92.0%	95.8%	+\$825,000 retained legitimate provider revenue
ERISA Appeals Spend	\$420,000/yr	< \$80,000/yr	\$52,000/yr	-\$368,000 sunk legal & clinical re-review spend

Direct Labor Savings: (18 FTEs - 5 FTEs) * \$78,888 = \$1,025,544 / Year
Appeals Cost Reduction: \$420,000 - \$52,000 = \$368,000 / Year
Prevented Denial Reimbursement: \$825,000 / Year | **AI Cloud Costs:** -\$35,280 / Year
Net Annual Operational Value: \$1,025,544 + \$368,000 + \$825,000 - \$35,280 = **\$1,850,000 Risk-Adjusted Net**

3. Strategic Alternatives Evaluated by Executive Committee

Alternative	Estimated Cost	Operational Risk	Executive Decision
Option A: Offshore BPO Scale-Up	\$1.8M / Year (Ongoing)	High error rates; violates CMS 72-hour turnaround mandate	REJECTED
Option B: Autonomous Direct AI Denials	\$650k CapEx	Catastrophic legal/ERISA risk; state regulatory fines	REJECTED
Option C: Deterministic RAG + HITL Gateway	\$1.2M CapEx / \$35k OpEx	Zero clinical liability; human-in-the-loop sign-off	APPROVED (Selected)

4. Capital Allocation & Formal Sign-Off Table

Role / Approver	Name & Title	Signature	Decision Date	Approval Status
Executive Sponsor	Dr. Karen Reynolds, Chief Medical Officer	K. Reynolds, MD	2026-03-15	APPROVED
P&L Budget Owner	Marcus Vance, VP Utilization Management (UMR)	M. Vance	2026-03-16	APPROVED
Enterprise Architecture	Siddharth Patel, VP Healthcare Technology	S. Patel	2026-03-18	APPROVED
Lead AI Product Manager	Abhinav Kumar, Lead AI PM	A. Kumar	2026-03-14	SUBMITTED